Mortality Equations for CONUS

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Data

We took the FIA mortality data subset, filtered out missing data (primarily crown ratio and heights). This left 4668763 observations. We found a number of observations with large decreases in crown closure at tree tip (`CCH`). For this analysis, we set the change in `CCH` to 0.0 for those observations by making the ending measurement equal to the initial measurement.

The observations are distributed among states is show in Table 1.

Table 1: Mortality Observations by State

State	n
MN	419443
ME	345773
WI	338255
MI	312582
OR	241139
GA	187003
NY	179379
PA	159036
VA	155130
WA	150603
AL	149441
NC	145877
MO	142716
CA	139743
SC	126350
MS	121309
AR	120221
TN	115117
MT	95025

Table 1: Mortality Observations by State

State	n
KY	92273
TX	87922
WV	83552
CO	79722
NH	78479
FL	64260
ID	61960
VT	55979
OH	55465
IN	44348
LA	44346
OK	38222
IL	32581
MA	26030
UT	24542
IA	18274
NM	17036
AZ	16740
NJ	15187
SD	14121
MD	14062
WY	11708
CT	11649
KS	11408
NE	7561
ND	5763
RI	5045
DE	4656
NV	1730

Base Rate Mortality

Daniel Kanneman and Amos Tversky[1] made the point that any putative model should improve upon the *base rate*. In our case, the base rate is the average mortality rate for each species accounting for no other factors. We calculate this by taking the annualized mean mortality rate for each species in the data set.

[1] Kahneman D. and Tversky, A. (1973). On the Psychology of Prediction. Psychology Review. 80(4), 237-251. doi: 10.1037/h0034747

The base rate estimates for 183 species are distributed as shown in Figure 1. Most base rate annual survival estimates are between 98 and 100%.

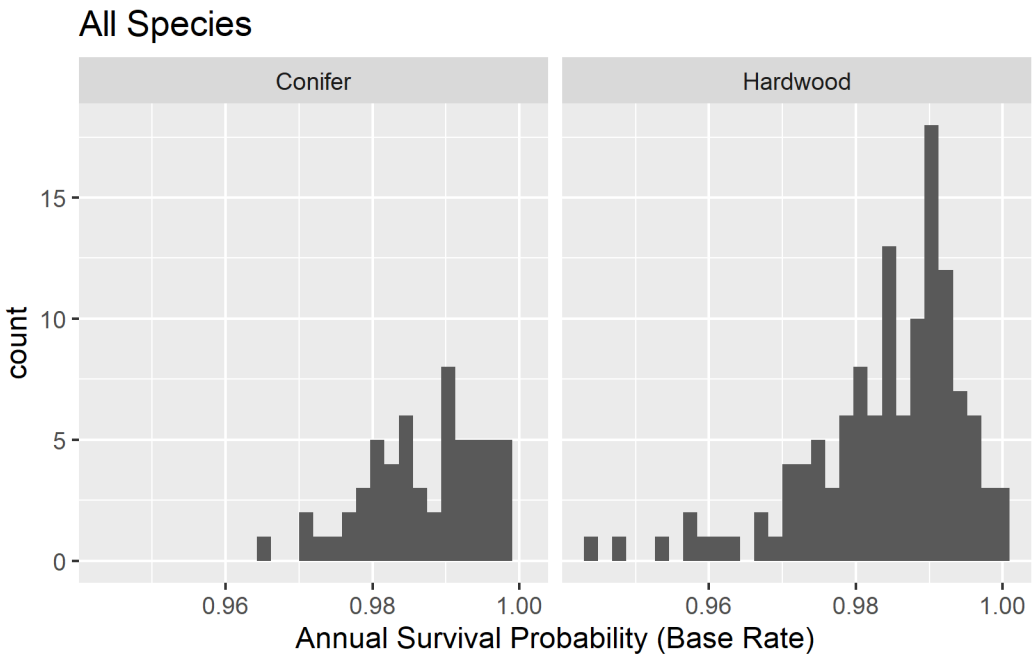
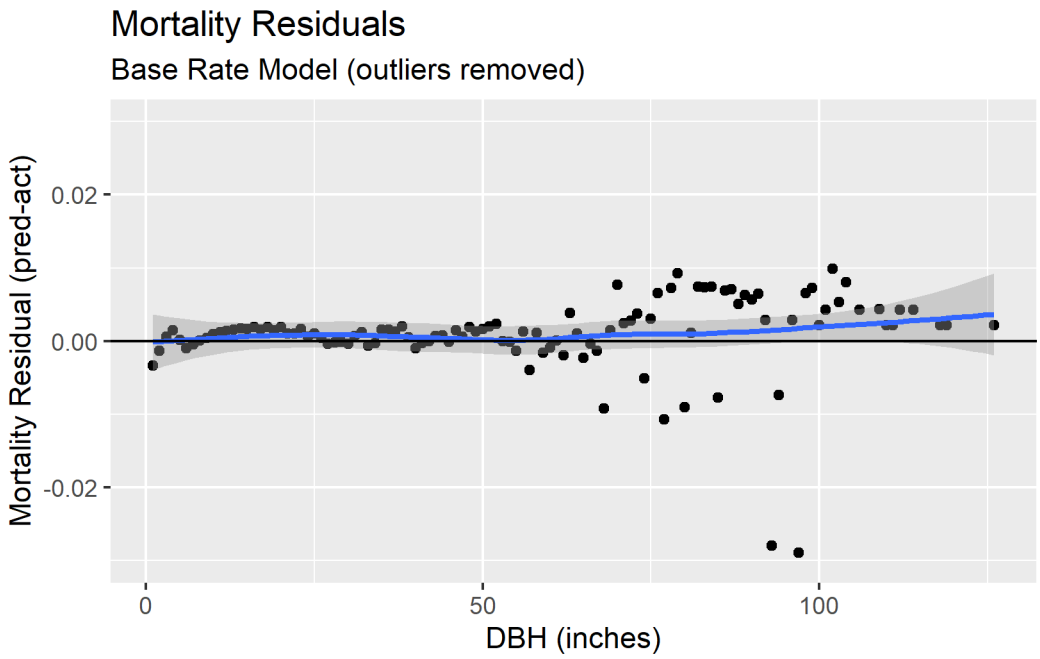


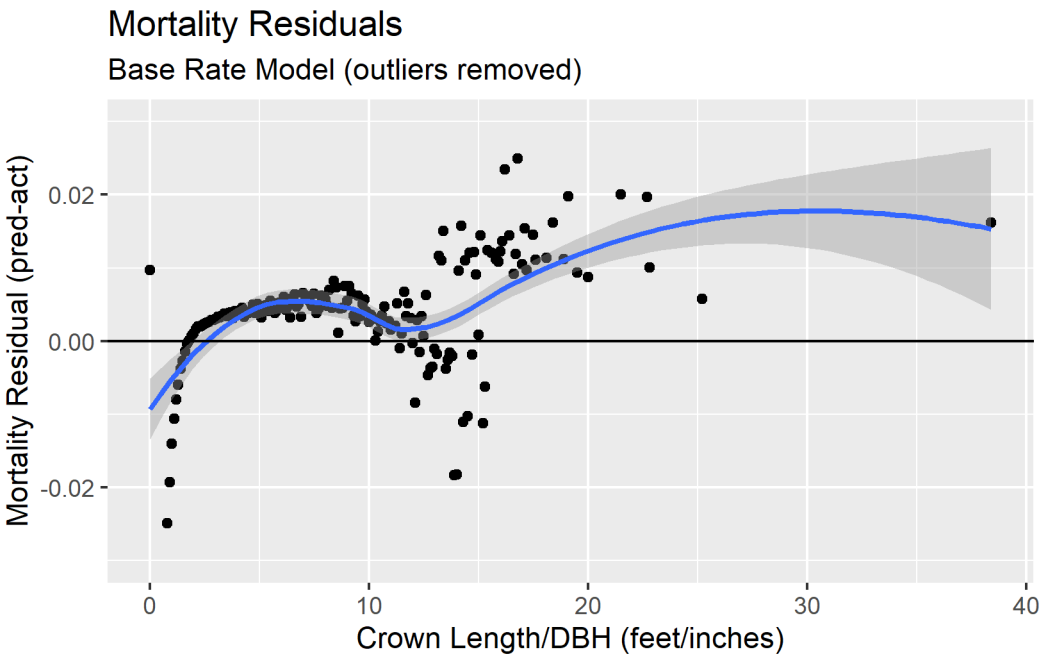
Figure 1

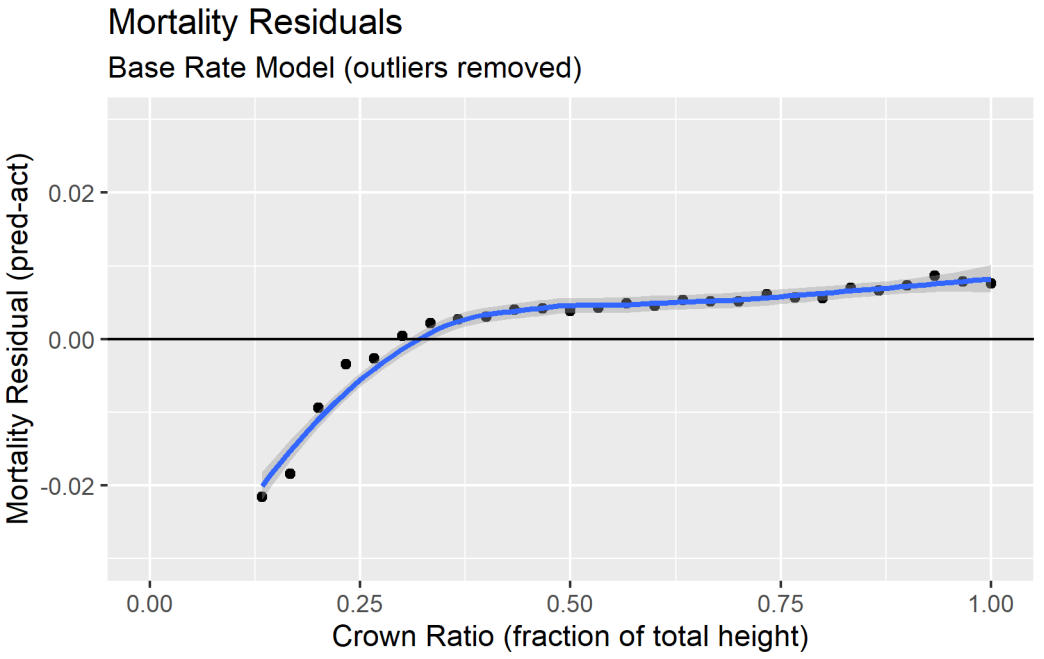
Base Rate Residuals

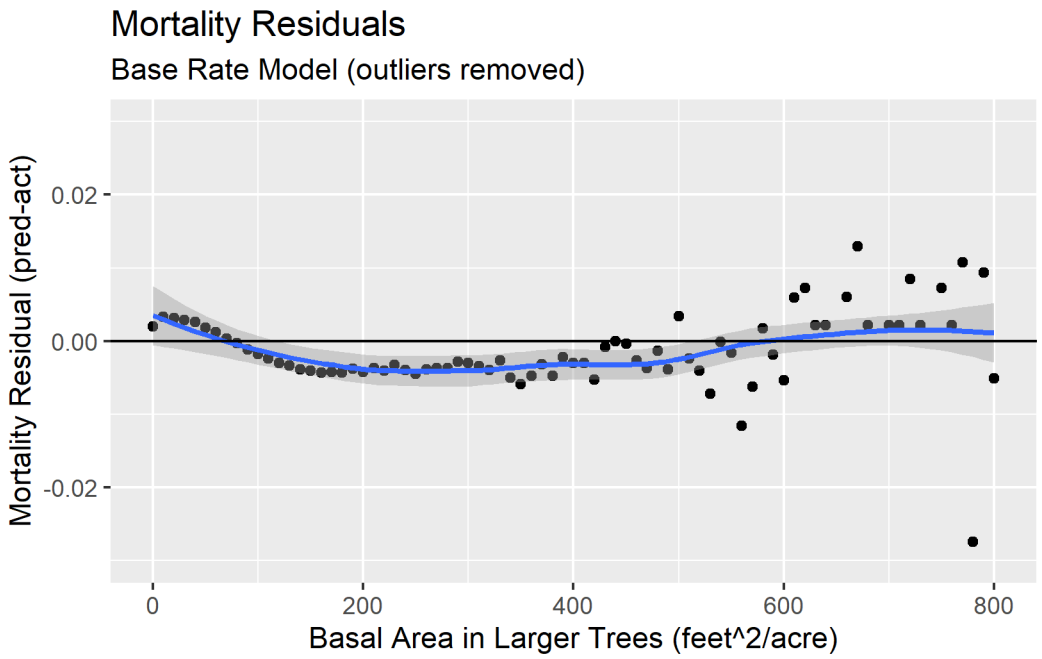
The following residual graphs show the average errors for all species over classes of possible explanatory variables.

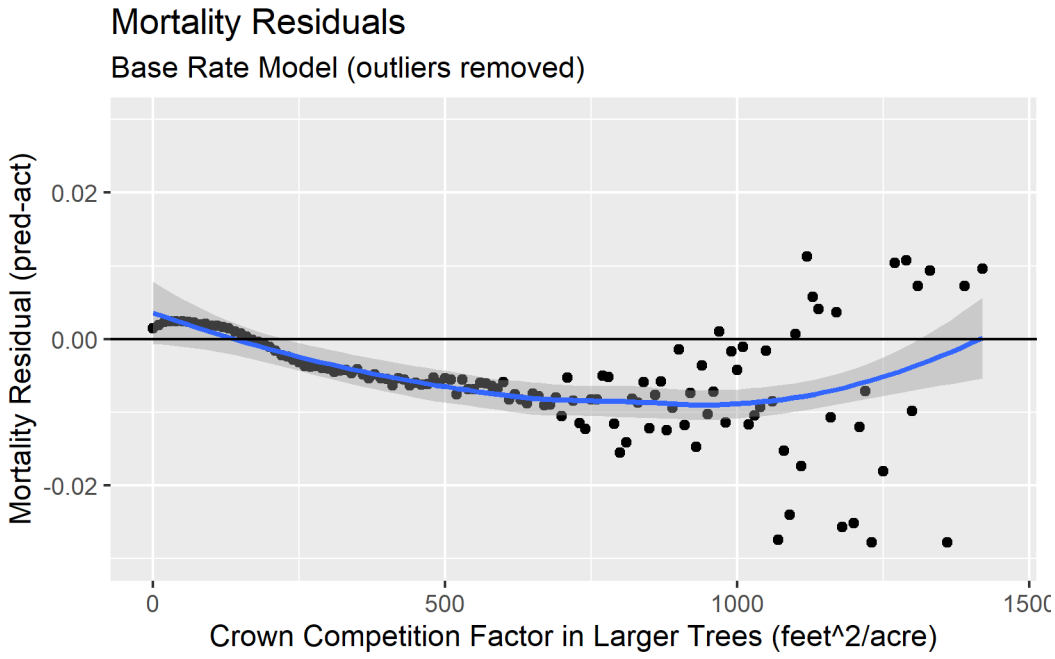

Mortality Residuals
Base Rate Model (outliers removed)
Mortality Residual (pred-act)
0.02
0.00
-0.02
0
50
100
DBH (inches)

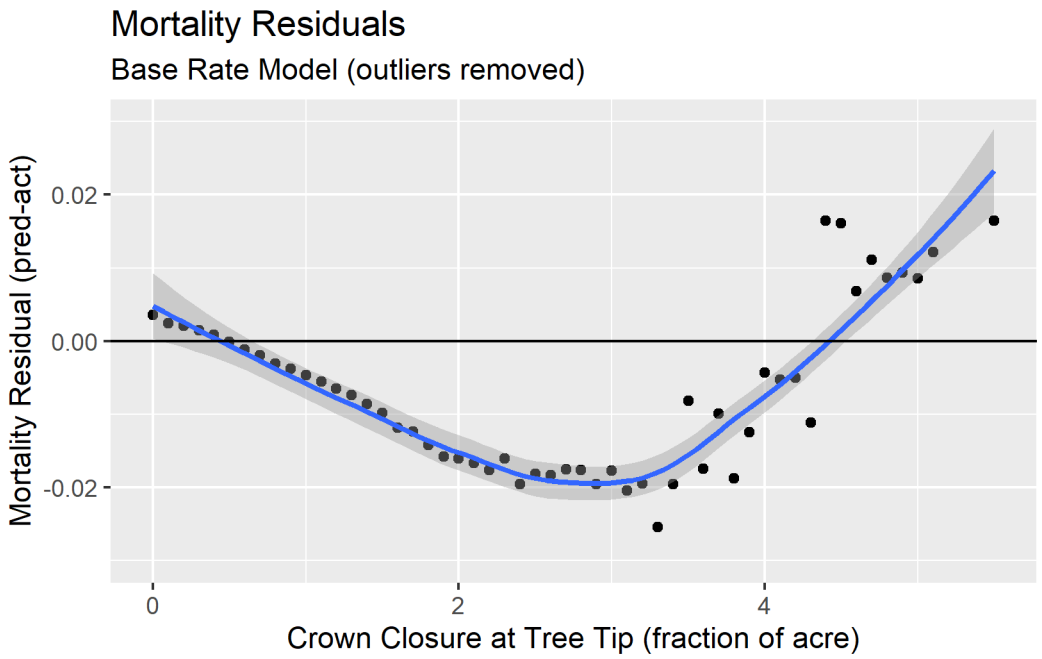

Mortality Residuals
Base Rate Model (outliers removed)
Mortality Residual (pred-act)
0.02
0.00
-0.02
0
100
200
300
Height (feet)

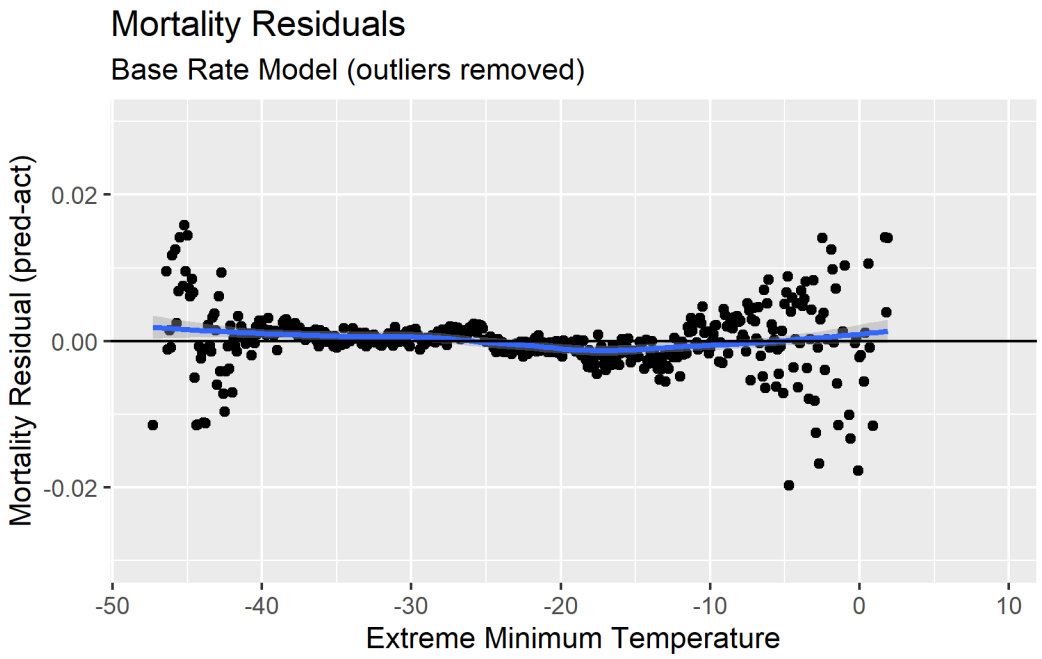

Mortality Residuals
Base Rate Model (outliers removed)
Mortality Residual (pred-act)
0.02
0.00
-0.02
0
10
20
30
40
Crown Length/DBH (feet/inches)

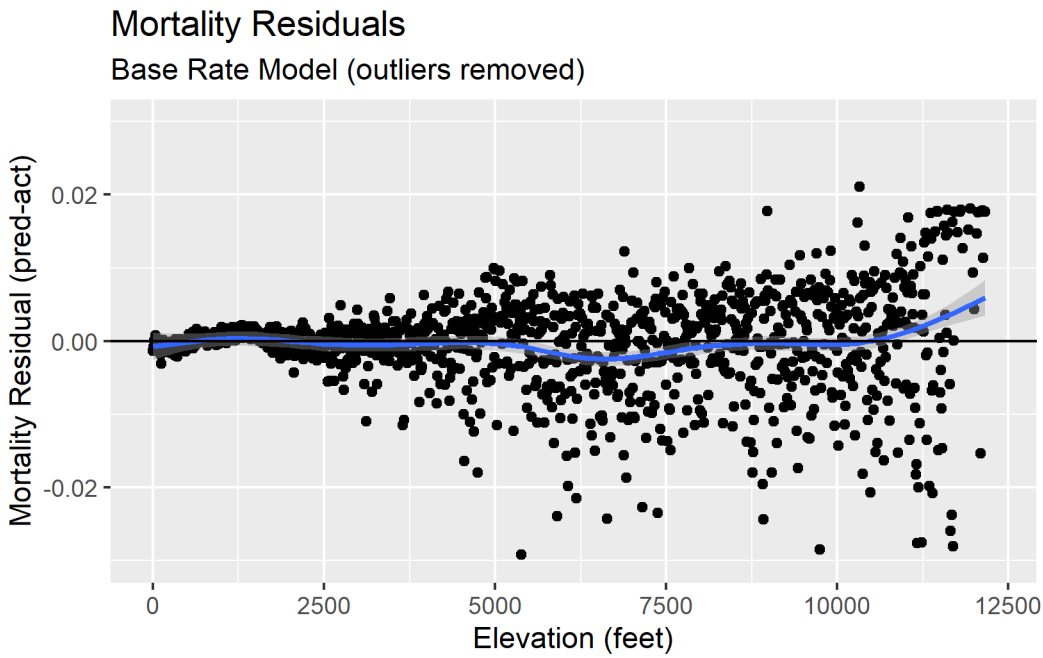

Mortality Residuals
Base Rate Model (outliers removed)
Mortality Residual (pred-act)
0.02
0.00
-0.02
0.00
0.25
0.50
0.75
1.00
Crown Ratio (fraction of total height)

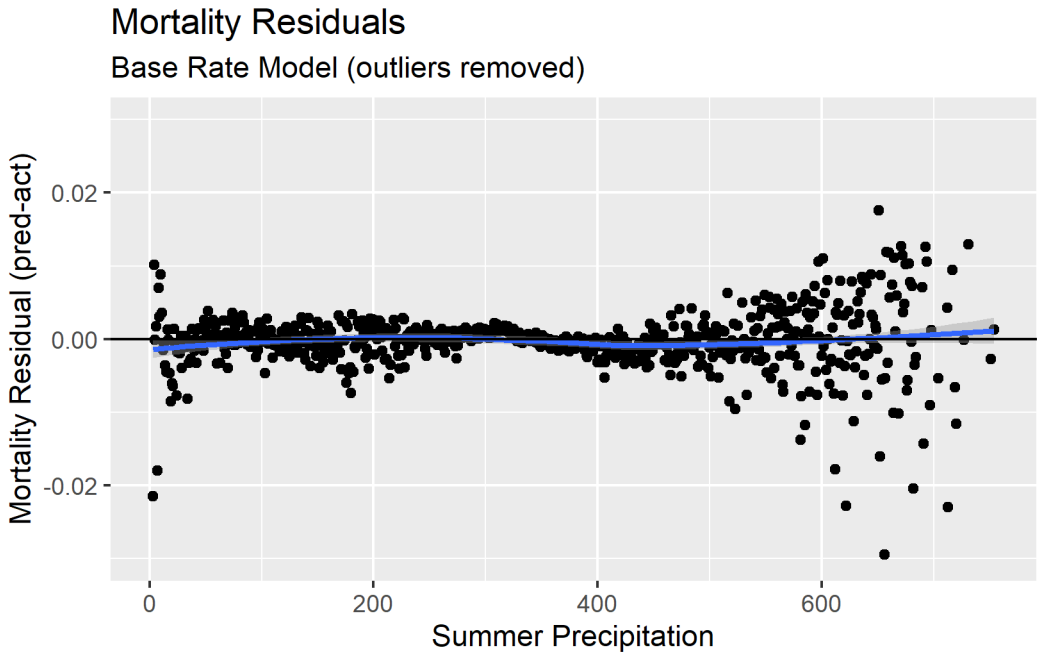

Mortality Residuals
Base Rate Model (outliers removed)
Mortality Residual (pred-act)
0.02
0.00
-0.02
0
200
400
600
800
Basal Area in Larger Trees (feet^2/acre)

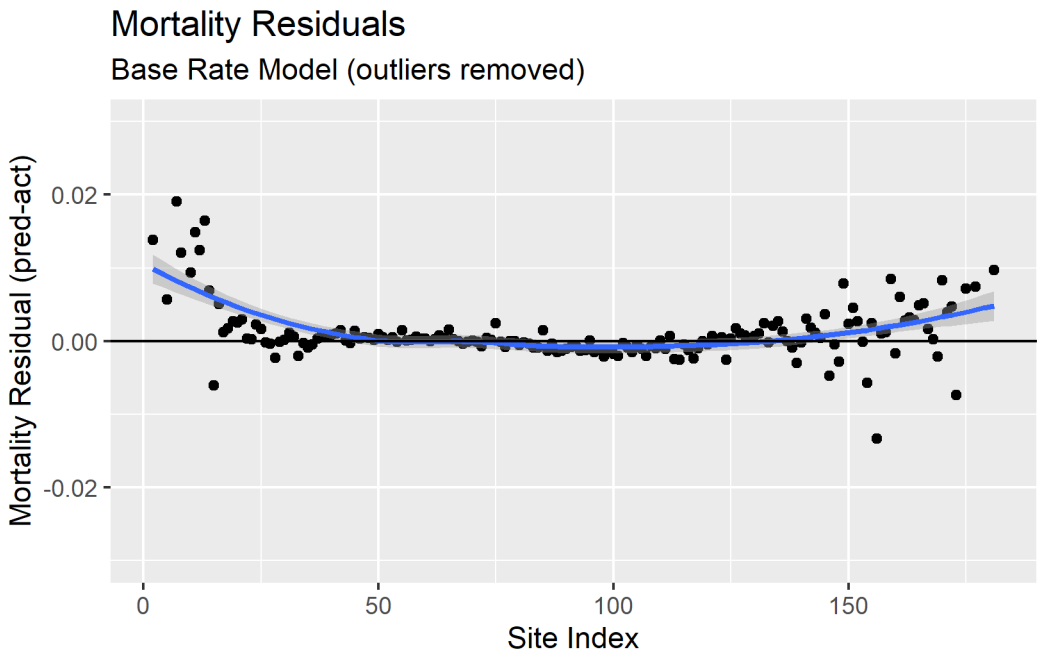

Mortality Residuals
Base Rate Model (outliers removed)
Mortality Residual (pred-act)
0.02
0.00
-0.02
0
500
1000
1500
Crown Competition Factor in Larger Trees (feet^2/acre)

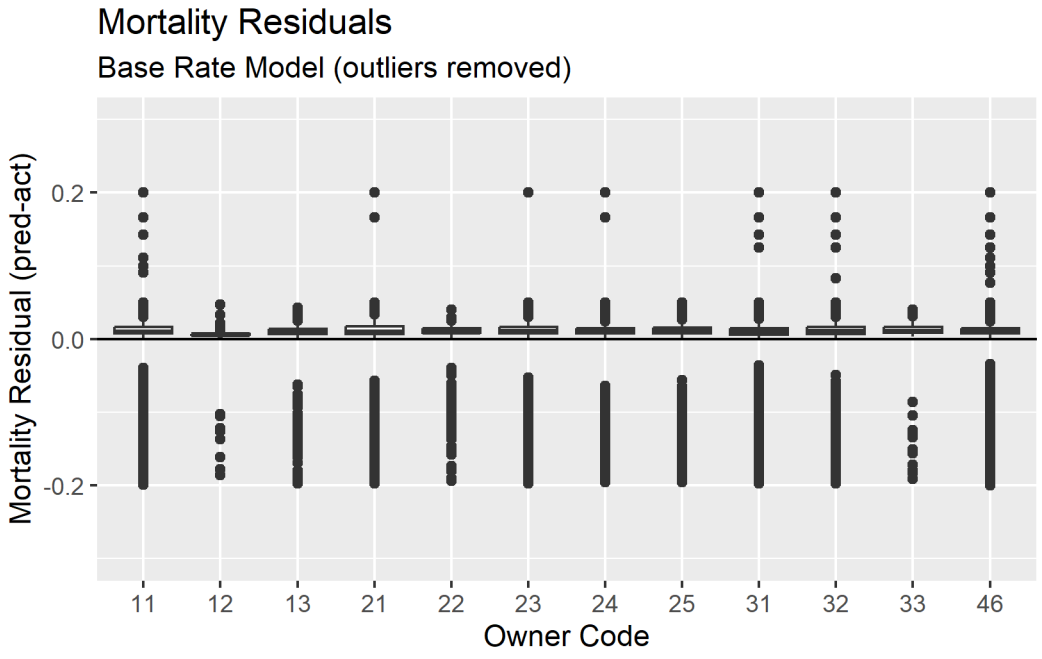

Mortality Residuals
Base Rate Model (outliers removed)
Mortality Residual (pred-act)
0.02
0.00
-0.02
0
2
4
Crown Closure at Tree Tip (fraction of acre)

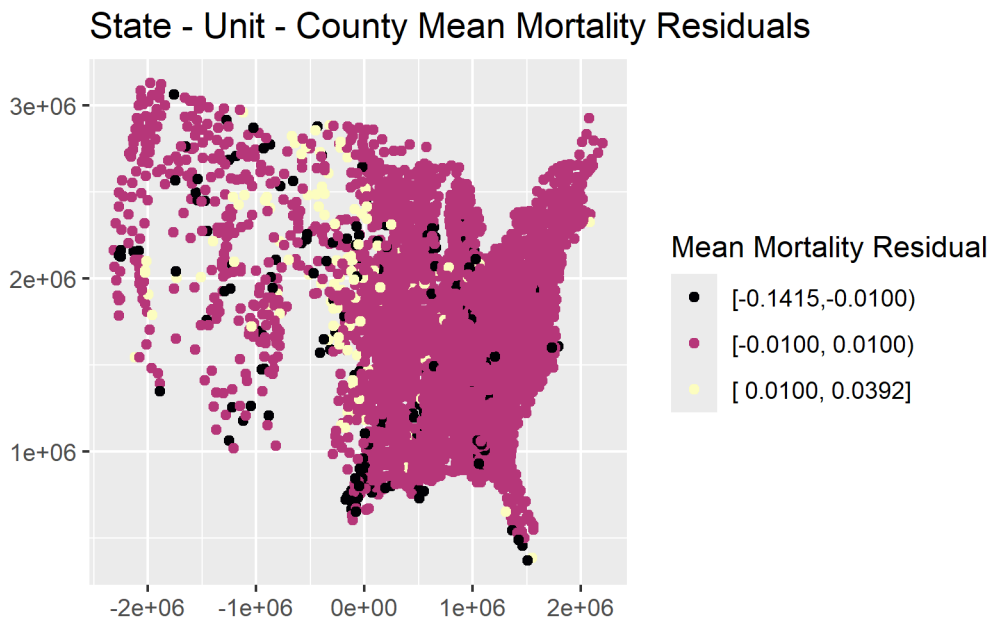

Mortality Residuals
Base Rate Model (outliers removed)
Mortality Residual (pred-act)
0.02
0.00
-0.02
-50
-40
-30
-20
-10
0
10
Extreme Minimum Temperature


Mortality Residuals
Base Rate Model (outliers removed)
Mortality Residual (pred-act)
0.02
0.00
-0.02
0
2500
5000
7500
10000
12500
Elevation (feet)


Mortality Residuals
Base Rate Model (outliers removed)
Mortality Residual (pred-act)
0.02
0.00
-0.02
0
200
400
600
Summer Precipitation


Mortality Residuals
Base Rate Model (outliers removed)
Mortality Residual (pred-act)
0.02
0.00
-0.02
0
50
100
150
Site Index


Mortality Residuals
Base Rate Model (outliers removed)
Mortality Residual (pred-act)
0.2
0.0
-0.2
11
12
13
21
22
23
24
25
31
32
33
46
Owner Code



Proposed Model

The model we propose to explain more of the mortality variation over the base rate is a Gompit function:

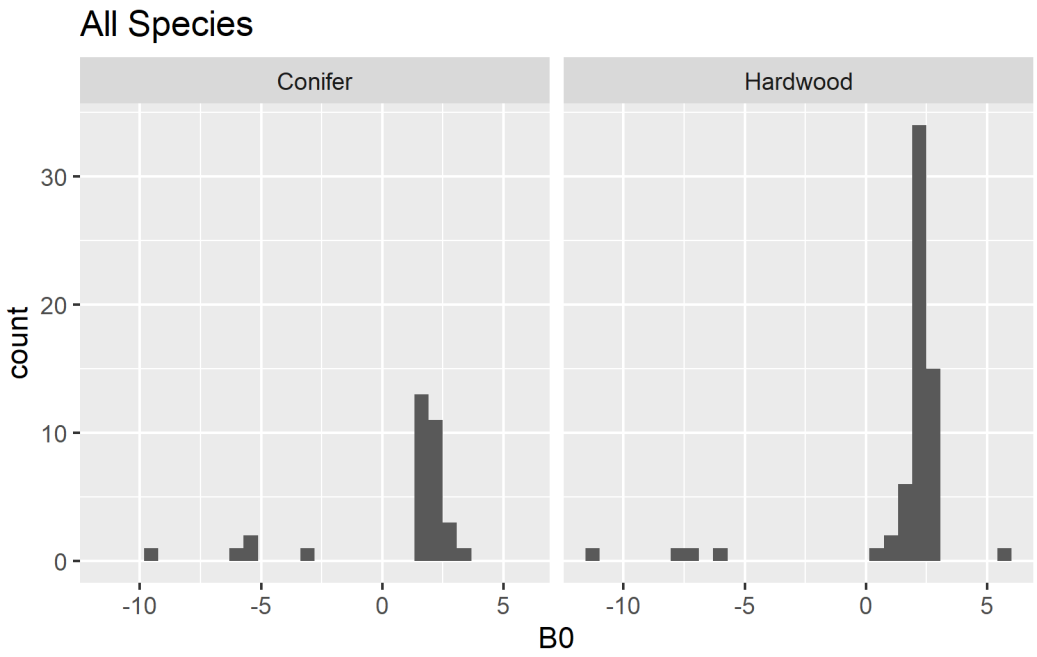
$$P_{surv} = 1 - e^{(-e^{(\beta_0+\beta_1(cr+0.01)^{\beta_2}+\beta_3 cch^{\beta_4})})} \tag{1}$$

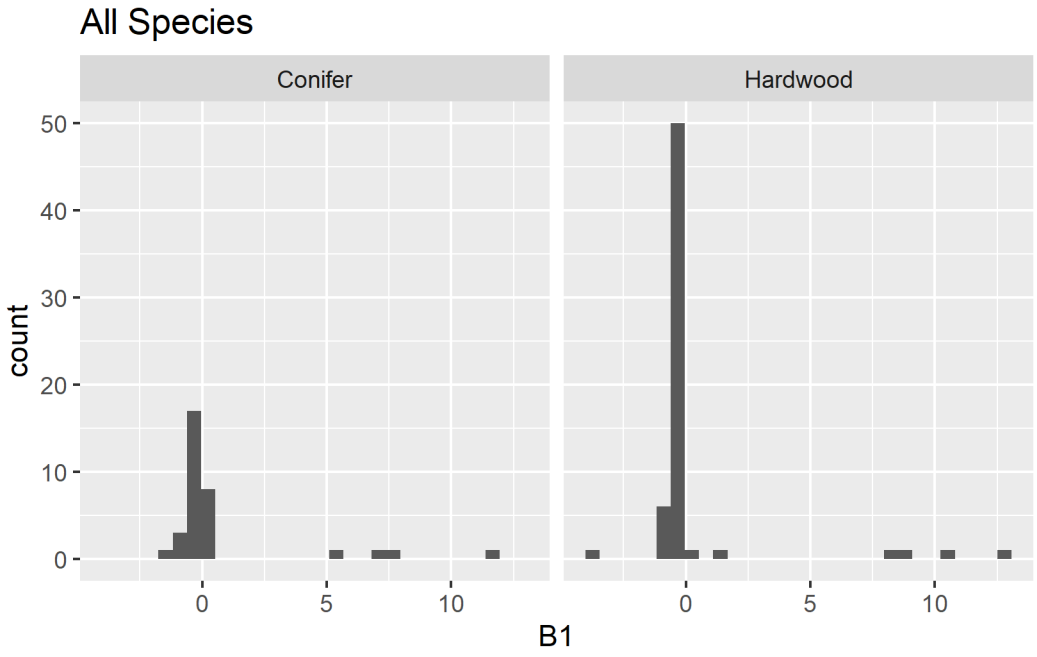
where:

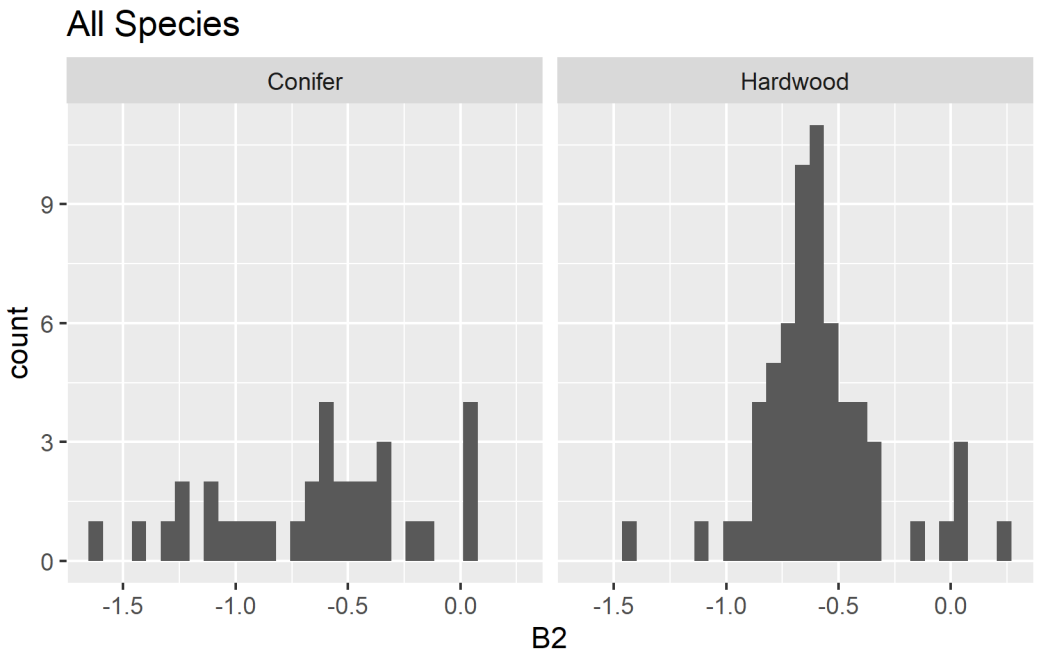
- P_{surv} : probability of survival
- cr : crown ratio
- cch : crown closure at tree tip

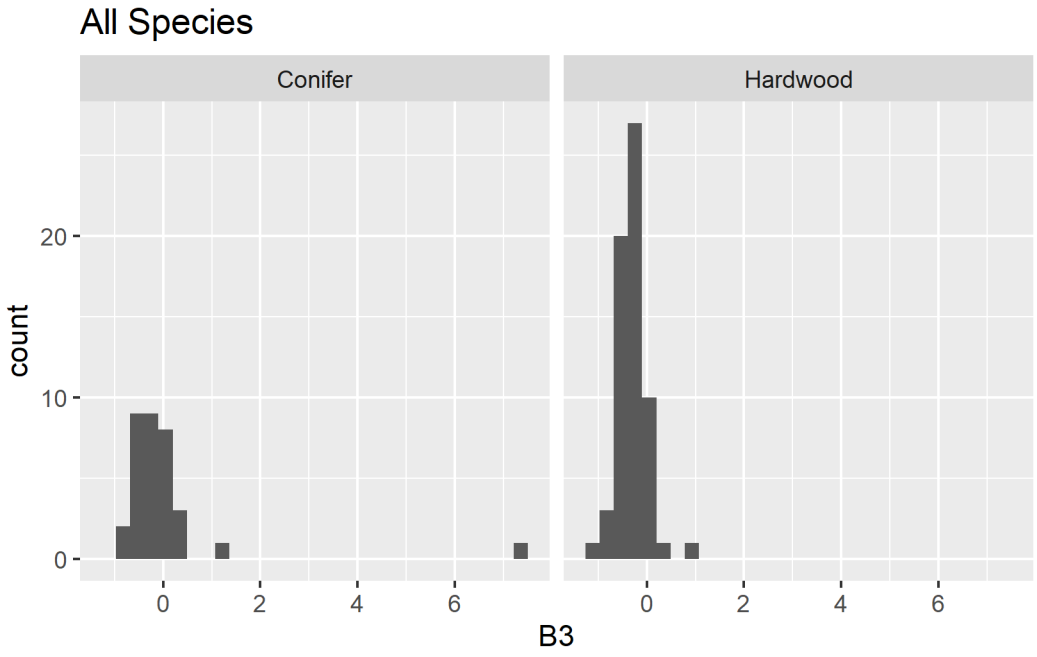
We used an integrative fitting approach to account for varying period lengths and we minimized the negative log-likelihood for each species separately.

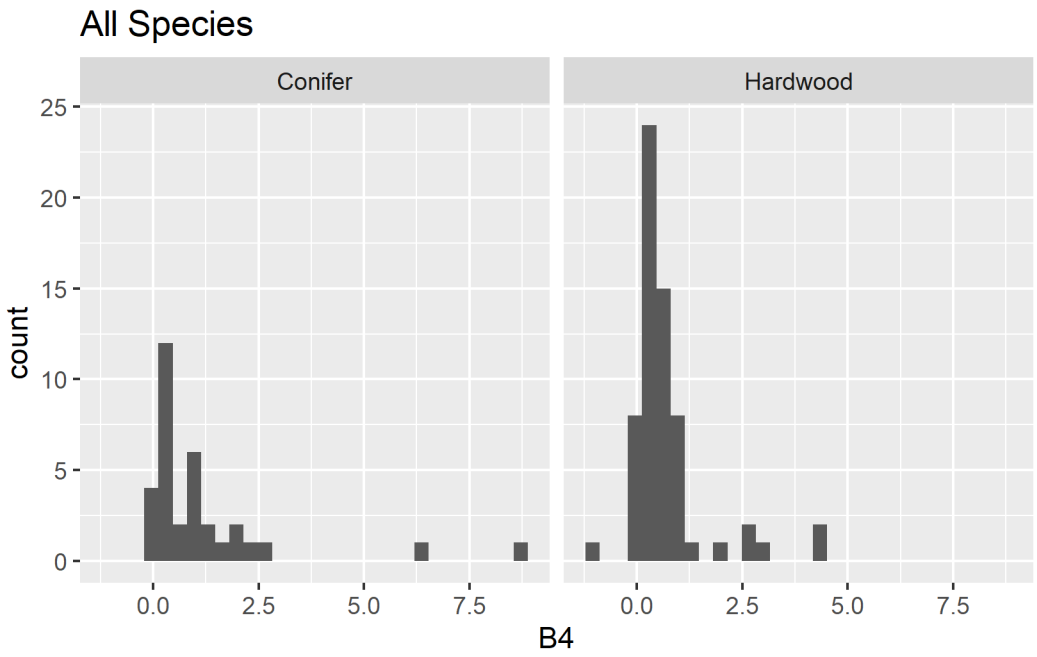
We were able to make parameter estimates for 96 species. The distributions of the estimates for these species are shown below.


All Species
Conifer
Hardwood
count
30
20
10
0
-10
-5
0
5
B0


All Species
Conifer
Hardwood
count
50
40
30
20
10
0
0
5
10
B1


All Species
Conifer
Hardwood
count
9
6
3
0
-1.5
-1.0
-0.5
0.0
B2


All Species
Conifer
Hardwood
count
20
10
0
0
2
4
6
B3



There are 9 species with parameter estimates that diverge significantly from the rest of the species. These species are listed in Table 2.

Table 2: Species with Extreme Parameter Estimates

FIA SPCD	Common Name	N Obs.
241	northern white-cedar	118180
110	shortleaf pine	48502
122	ponderosa pine	97875
832	chestnut oak	59831
132	Virginia pine	26229
971	winged elm	23855
602	black walnut	12591
242	western redcedar	16891
805	canyon live oak	7662

Figure 2 shows that all species' negative log likelihoods were reduced over the base rate with the proposed model.

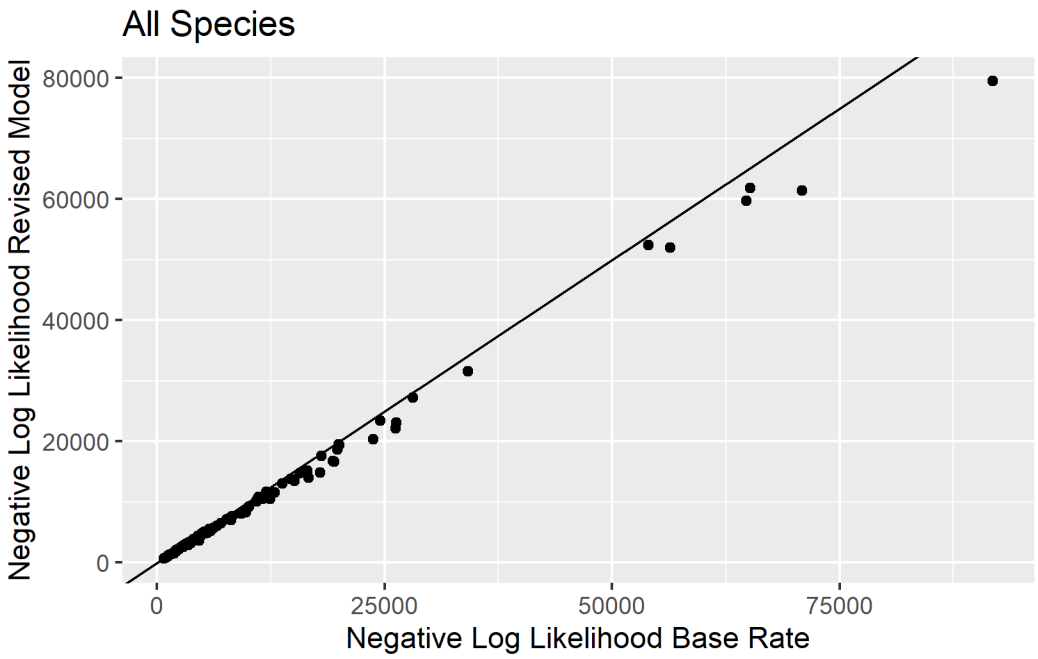
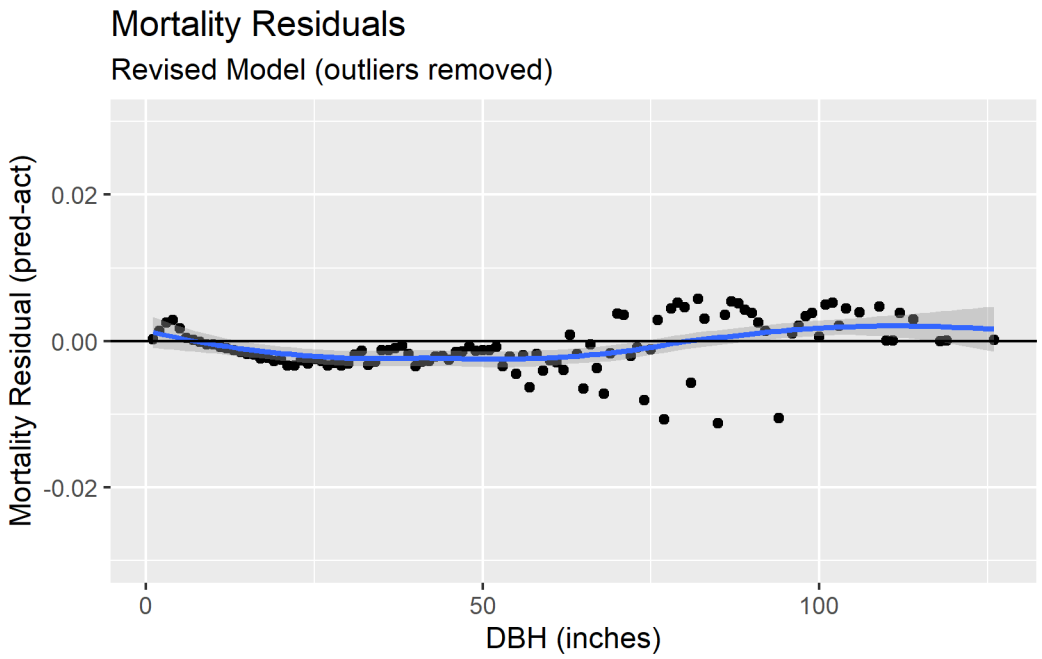


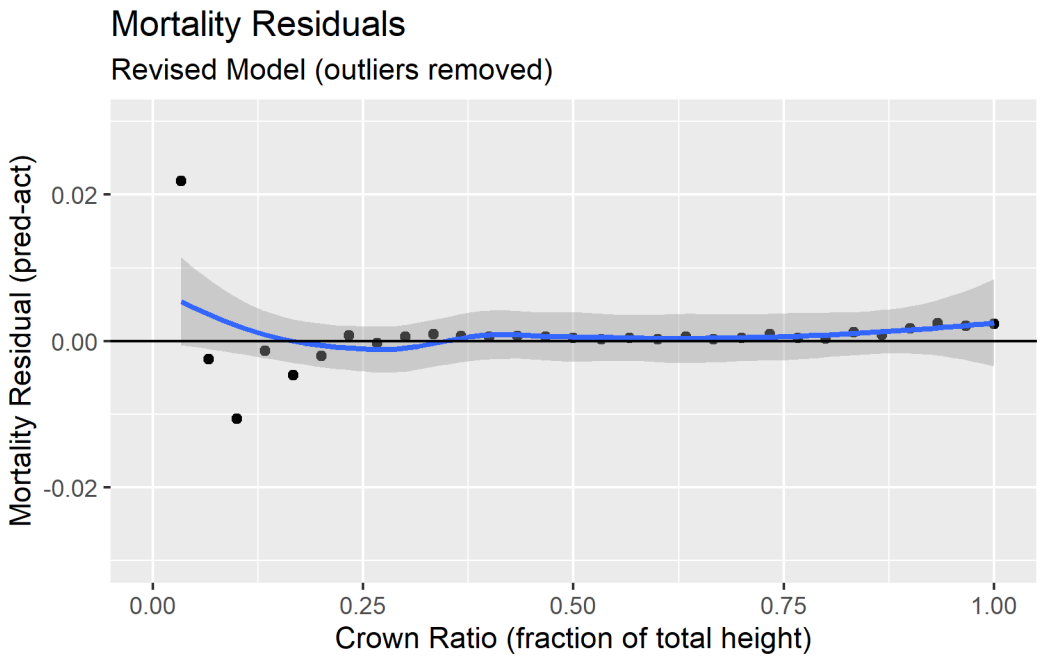
Figure 2

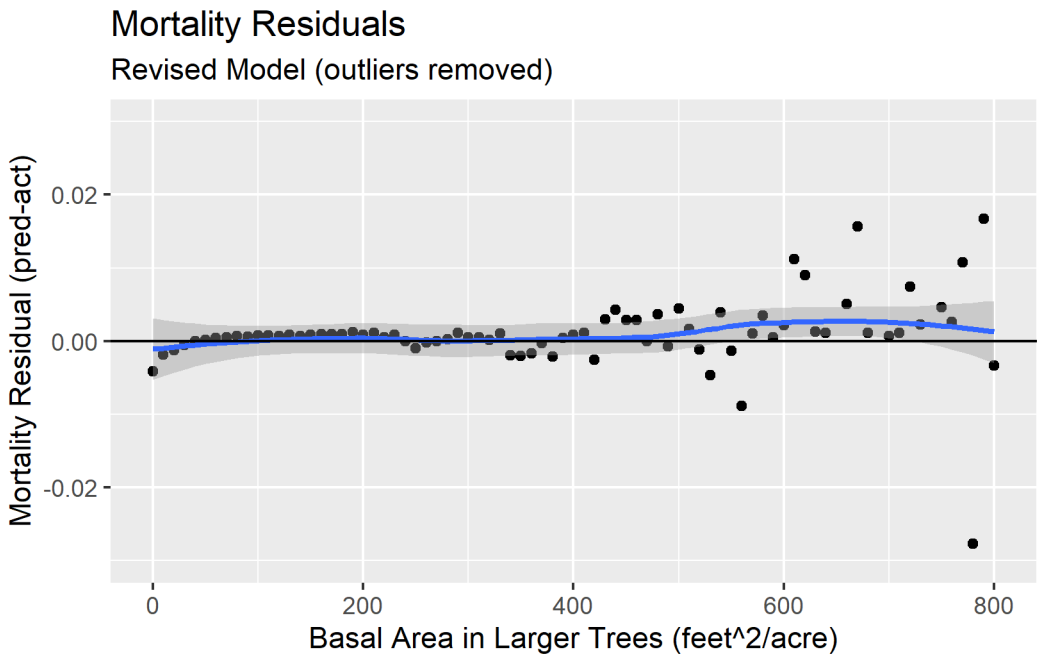
Proposed Model Residuals

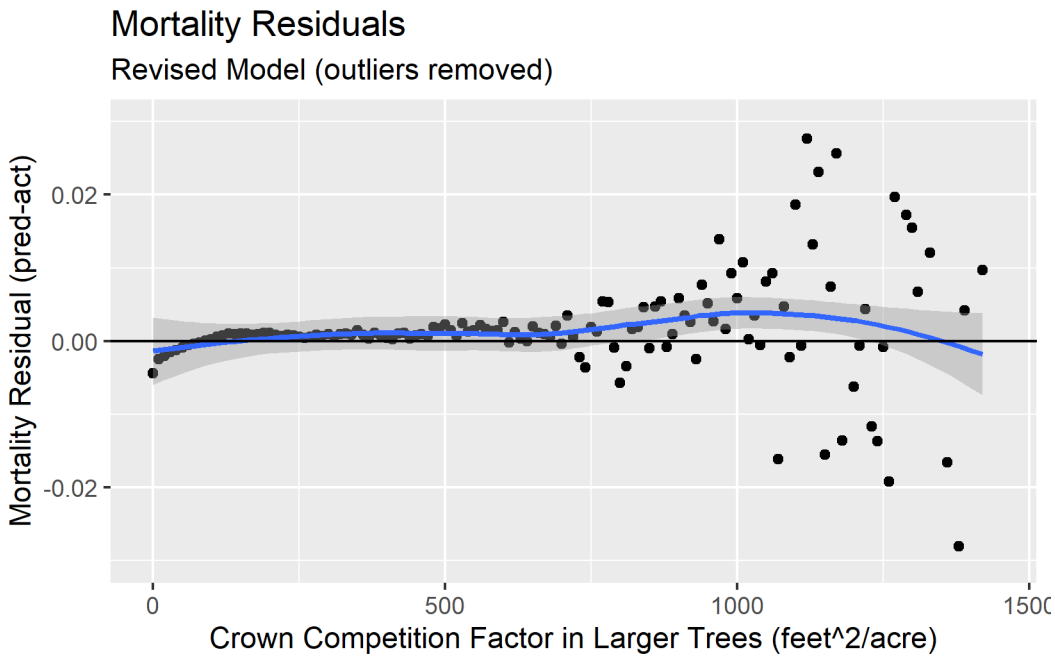
The following residual graphs show the average errors for all species where the fit improved over the base rate over classes of possible explanatory variables.

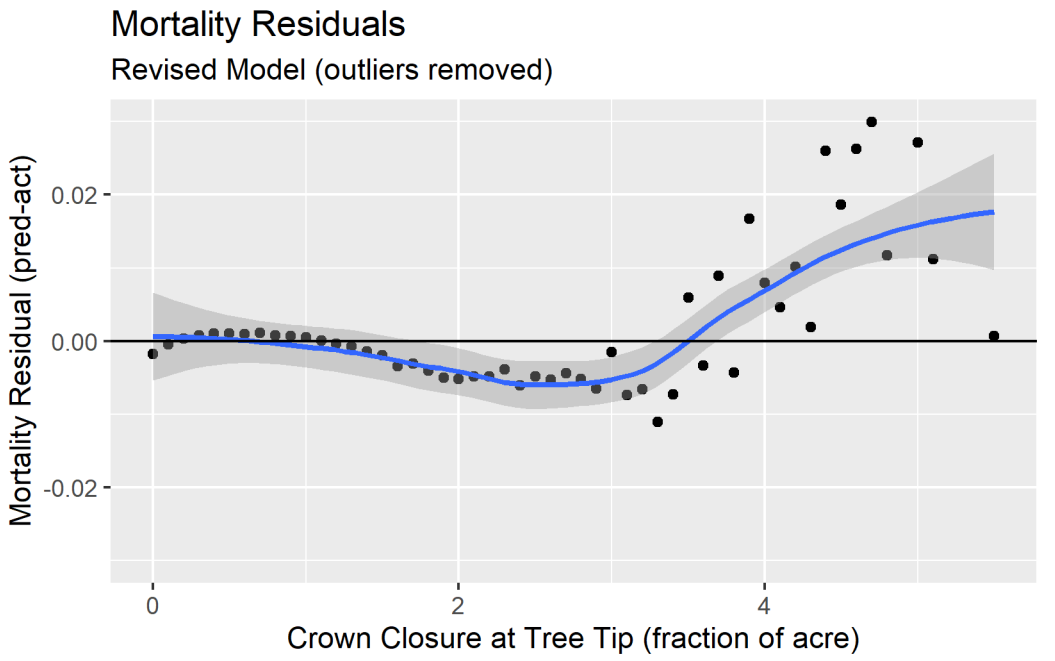

Mortality Residuals
Revised Model (outliers removed)
Mortality Residual (pred-act)
0.02
0.00
-0.02
0
50
100
DBH (inches)

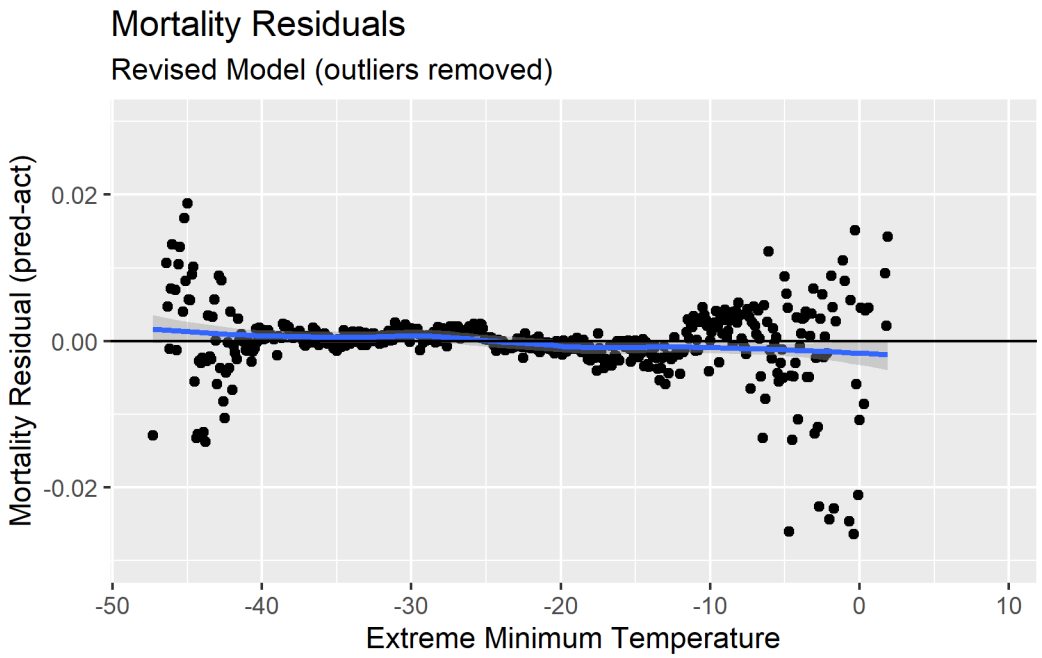

Mortality Residuals
Revised Model (outliers removed)
Mortality Residual (pred-act)
0.02
0.00
-0.02
0
100
200
300
Height (feet)

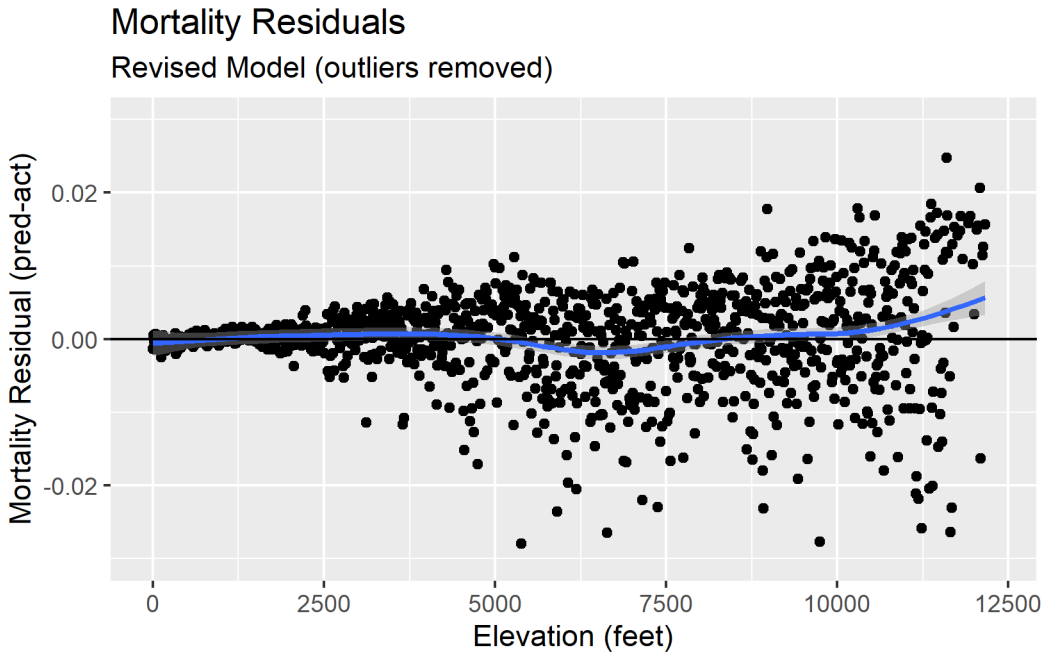

Mortality Residuals
Revised Model (outliers removed)
Mortality Residual (pred-act)
0.02
0.00
-0.02
0.00
0.25
0.50
0.75
1.00
Crown Ratio (fraction of total height)

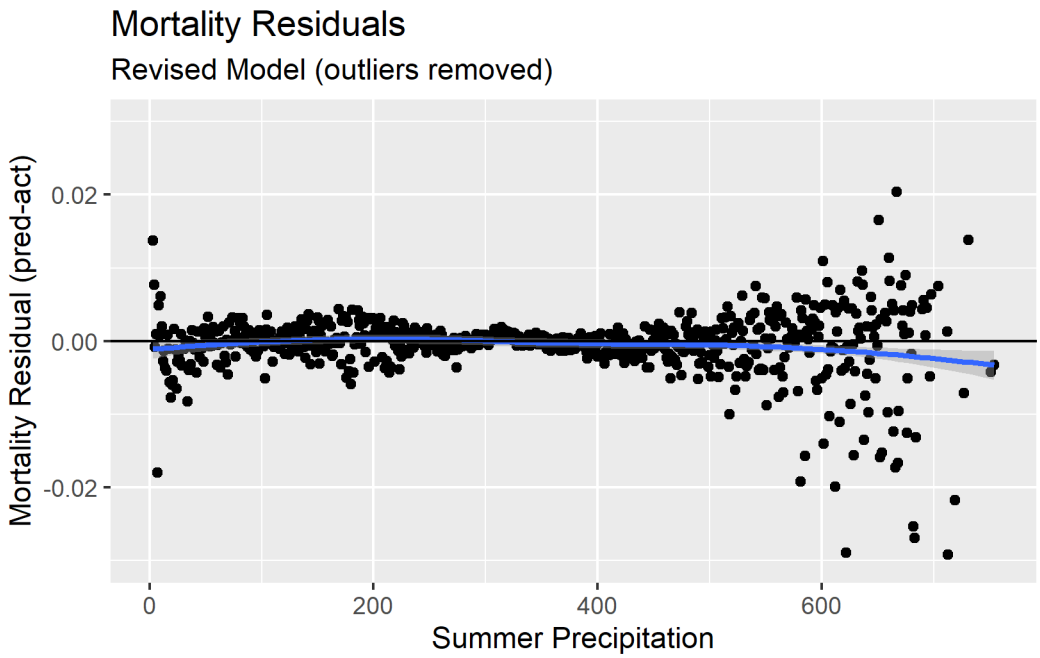

Mortality Residuals
Revised Model (outliers removed)
Mortality Residual (pred-act)
0.02
0.00
-0.02
0
200
400
600
800
Basal Area in Larger Trees (feet^2/acre)


Mortality Residuals
Revised Model (outliers removed)
Mortality Residual (pred-act)
0.02
0.00
-0.02
0
500
1000
1500
Crown Competition Factor in Larger Trees (feet^2/acre)

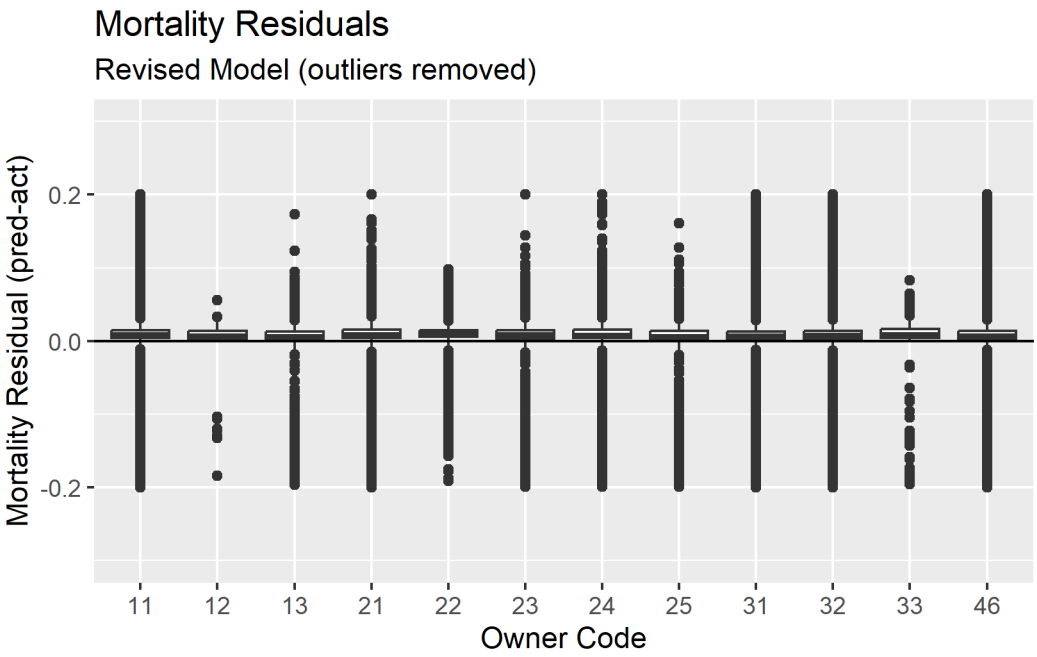

Mortality Residuals
Revised Model (outliers removed)
Mortality Residual (pred-act)
0.02
0.00
-0.02
0
2
4
Crown Closure at Tree Tip (fraction of acre)

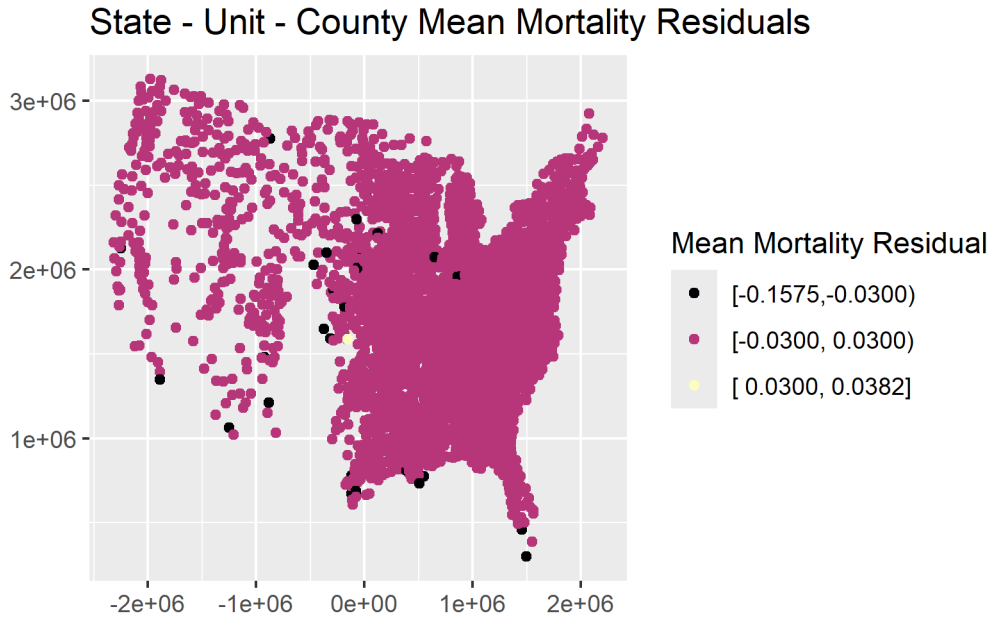

Mortality Residuals
Revised Model (outliers removed)
Mortality Residual (pred-act)
0.02
0.00
-0.02
-50
-40
-30
-20
-10
0
10
Extreme Minimum Temperature


Mortality Residuals
Revised Model (outliers removed)
Mortality Residual (pred-act)
0.02
0.00
-0.02
0
2500
5000
7500
10000
12500
Elevation (feet)


Mortality Residuals
Revised Model (outliers removed)
Mortality Residual (pred-act)
0.02
0.00
-0.02
0
200
400
600
Summer Precipitation


Mortality Residuals
Revised Model (outliers removed)
Mortality Residual (pred-act)
0.02
0.00
-0.02
0
50
100
150
Site Index


Mortality Residuals
Revised Model (outliers removed)
Mortality Residual (pred-act)
0.2
0.0
-0.2
11
12
13
21
22
23
24
25
31
32
33
46
Owner Code


State - Unit - County Mean Mortality Residuals
3e+06
2e+06
1e+06
-2e+06
-1e+06
0e+00
1e+06
2e+06
Mean Mortality Residual
[-0.1575,-0.0300)
[-0.0300, 0.0300)
[0.0300, 0.0382]